

Programming of cRIO/FPGA system for High Temperature Superconductors (HTS) Magnets application

ALEJANDRO DIAZ FONTALVA

SUPERVISORS: JUAN CARLOS PEREZ AND HUGO BAJAS

CERN TE-MSC-MDT & TE-MSC-TF

Abstract

Working under the frame of the HTS₁ Project, I have been in charge of the Software implementation of the cRIO DAQ₂ system that will be use during the cold powering tests campaigns of different set-ups of HTS magnets. The cRIO DAQ system will be used in the future, during the cold powering tests campaigns of the HTS magnets, Feather M0.5 and Feather M2. Furthermore, I collaborated during the last two cold powering tests campaigns of the HTS coil Feather M0.4. These ones took place from August to the beginning of October of 2016. My tasks were helping with the data analysis of the measurements taken during the tests as well as fixing and improving the posible Hardware or Software problems found in the original cRIO DAQ system [1].

HTS Project & Feather M0.4 set-up

The main goal of the HTS project is the development of High Temperature Superconductors and their application for magnet production. The main unknown of the project is learn how to protect HTS magnets after a quench₃.

Feather M0.4

For cooling down tests have been used different set-ups and the one I've been involved is **Feather M0.4** [2] (Fig 1). This one consist of a coil wound by means of Roebel cable compounds by the HTS material. The coil has several kind of sensors placed along its surface (voltage taps, pick-up coils, hall probes, etc...) to monitor its behavior. Furthermore, there are installed 3 SH₄ to induce quenches in the coil. Also for state monitoring and quench protection are running 4 different DAQ system in parallel and the one I'm in charge is the cRIO DAQ system.

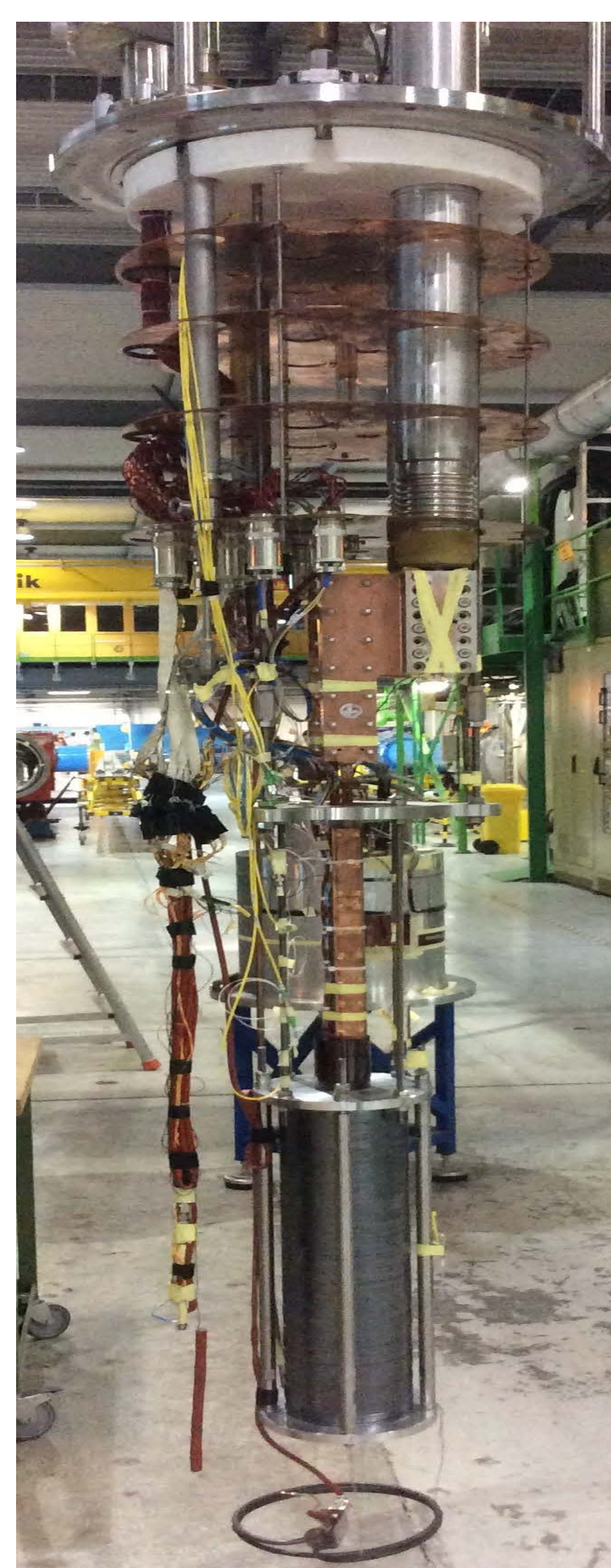


Fig 1: Feather M0.4 Set-up

Test Campaigns

Since July until the beginning of October 2016 were carried on 4 Cold Powering Test Campaigns or Runs over Feather M0.4 [3]. My personal contribution to the HTS team was during Runs 3 and 4.

Run 3

Procedure: Cooling down the coil to temperatures in the order of 20 to 80K. After this, the coil is powered increasing the current level until a quench is detected by some of the DAQ systems. In case a quench will be detected the magnet protections will shut down the powering. Finally the measured data are analyzed in the search of patterns which help us to predict when a quench will happen. (Fig 2).

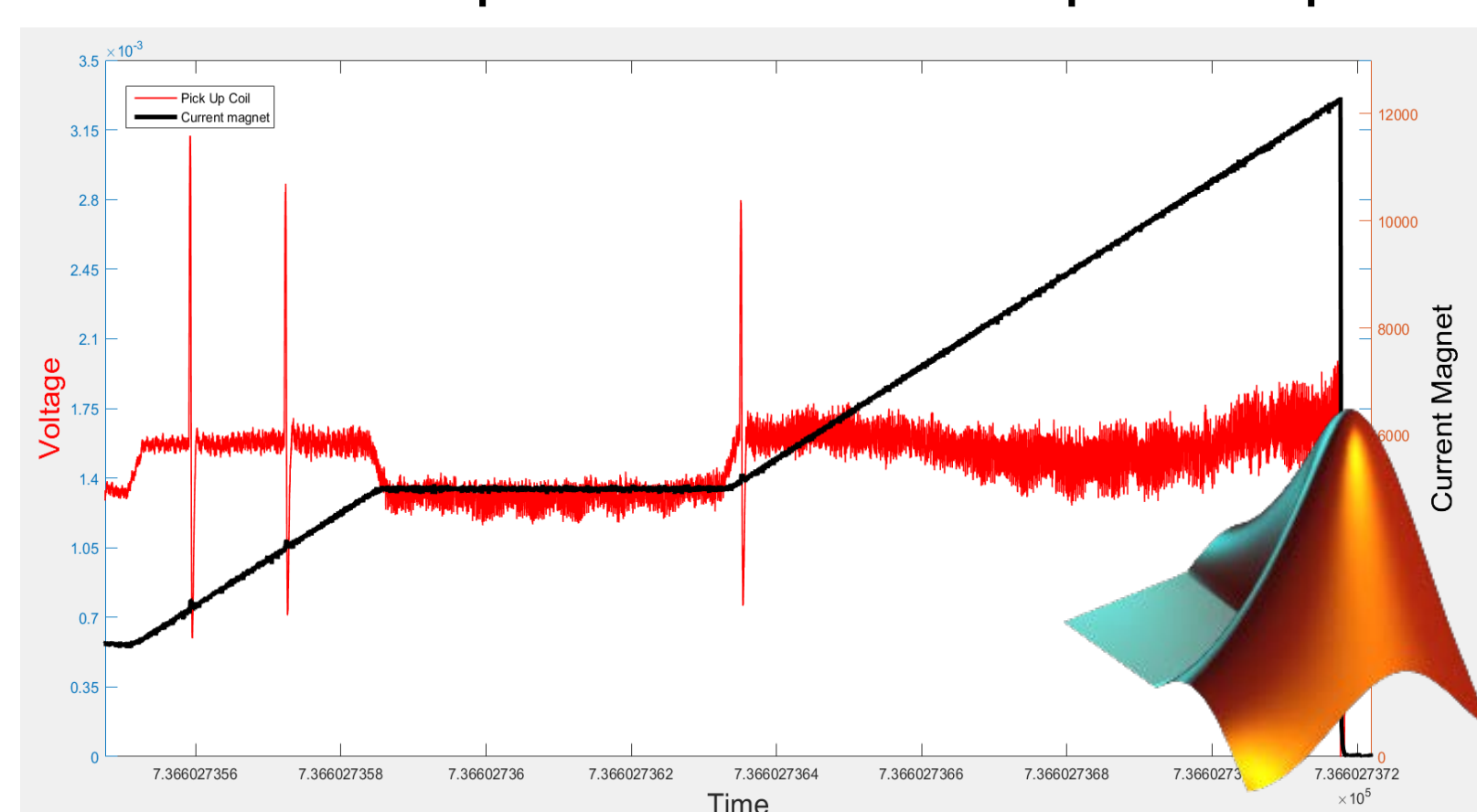


Fig 2: Quench at nominal current 12kA

Some problems related with the synchronization of the data of the DAQ systems and an oversized sample rate happened during Run 3. In order to solve these problems I made some improvements in the Matlab scripts used for the data analysis.

Run 4

Procedure: Unlike Run 3, in this case quenches were induced by means of powering pulses sent to SHs placed into the coil surface by means of the cRIO DAQ system. So that, firstly a current level target is preset. Once this current target is reached then any of the SH is fired. In case the magnet protection won't trigger then the width of the SH powering pulse is increased, else the magnet powering will be shutted down. During Run 4 were found multiples problems related with the SH pulses. In order to solved them it was calibrated again the Power Supply used to generate the powering pulses. Also I applied some conversion factors in the Matlab scripts used for the data analysis.

cRIO Software Implementation

cRIO [4] is a NI₅ device compound by means of an FPGA₆ and a RT₇ Processor as well as a chassis with 8 module slots for reconfigurable I/O. The communication with the cRIO is done by means of an IP through Ethernet connection. The current set-up used for the cRIO (Fig. 3) is provided with 7 modules of 32 Analog inputs and 1 module with 8 Digital Outputs. Furthermore, it has a 3Tb HD is used in order to store the measured data.

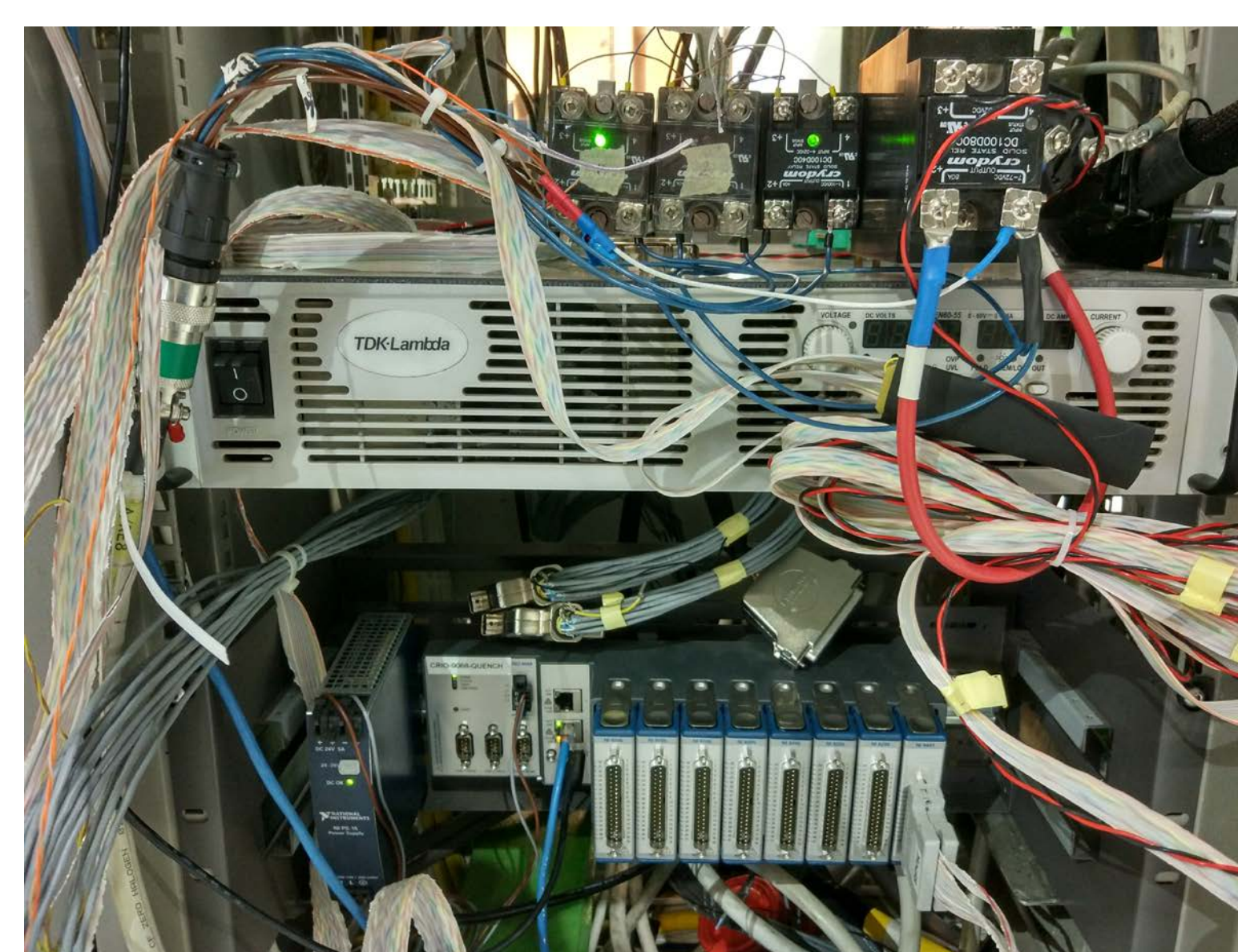


Fig 3: Current set-up of cRIO

The main tasks performed by the software programmed in the cRIO are Data Acquisition and Real time monitoring, the Spot Heater Firing System and the Quench Detection System. The software implementation is developed by LabVIEW software of NI and this one is structured in 3 main modules the FPGA, the RT Processor and the Host. (Fig. 4). Firstly the FPGA is in charge of the Readout of the sensors, the SH Firing system and the Quench Detection system.

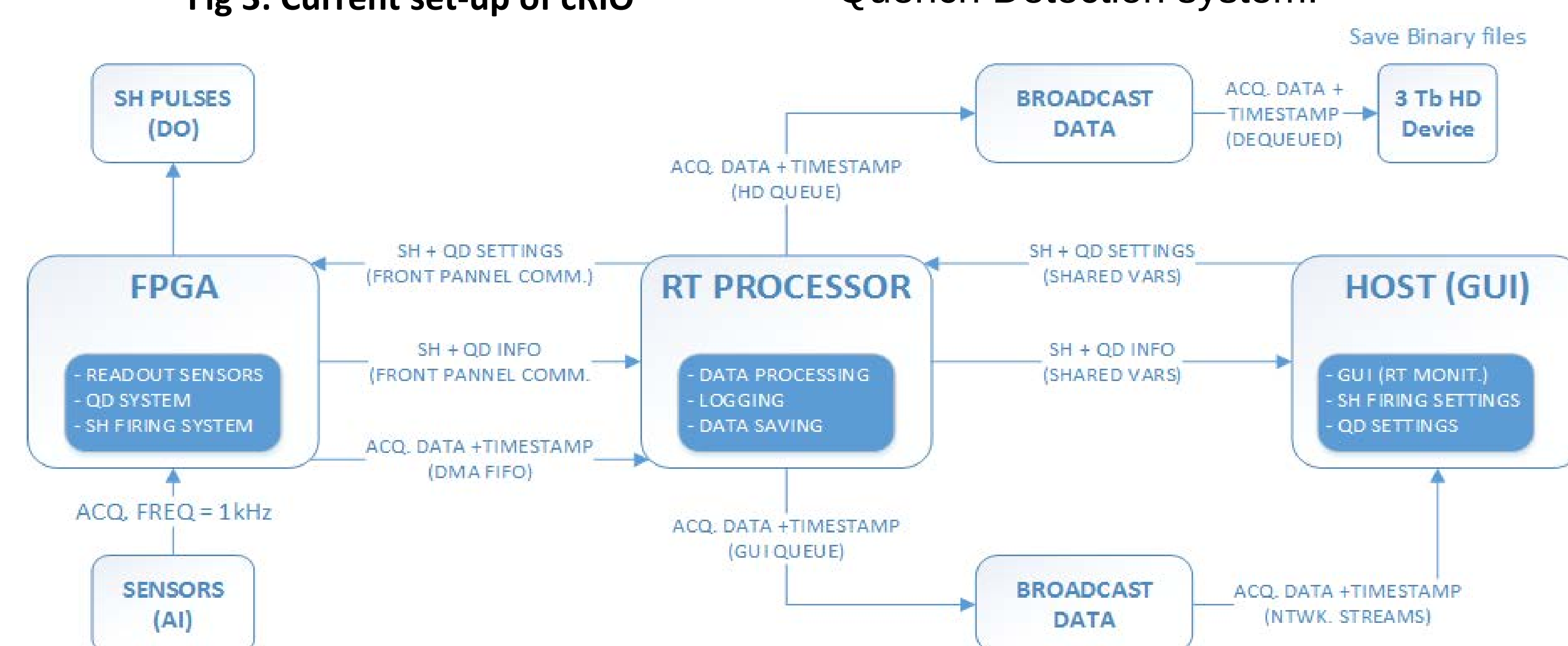


Fig 4: Flowchart Diagram of cRIO Software

On the other hand the RT Processor performs the data processing, the logging, and the storage of the data as well. Finally, the Host acts as the Graphical User Interface (GUI) (Fig. 5). Their features are the RT monitoring of the acquired signals, as well as modify the settings of the SH Firing or the Quench Detection system. The communication between the modules is performed by means of different LabVIEW communication protocols such as DMA FIFOs, queues, network streams, Shared variables or Front Panel communication.

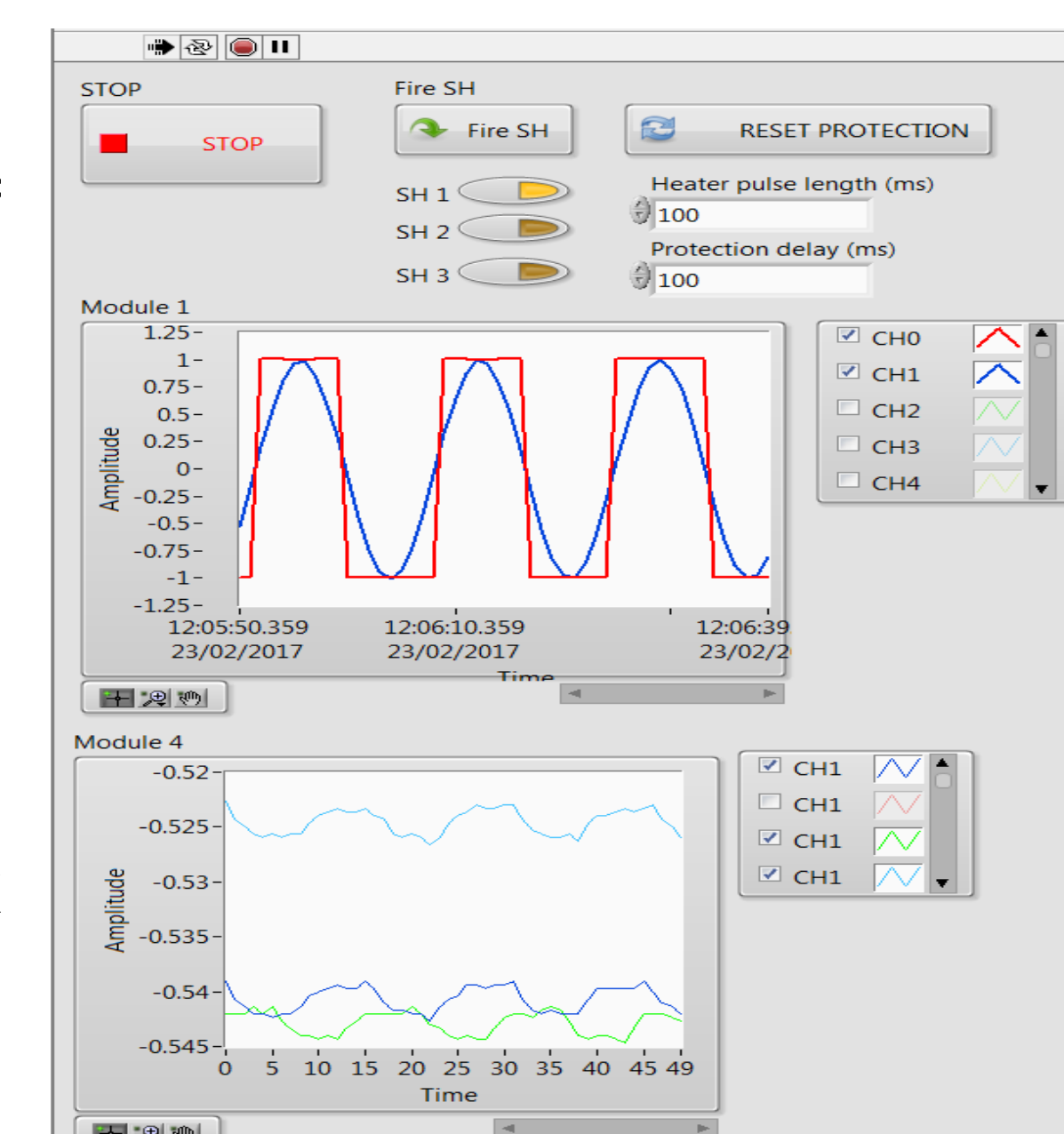


Fig 5: GUI of cRIO Software

Results & Current status of the Project

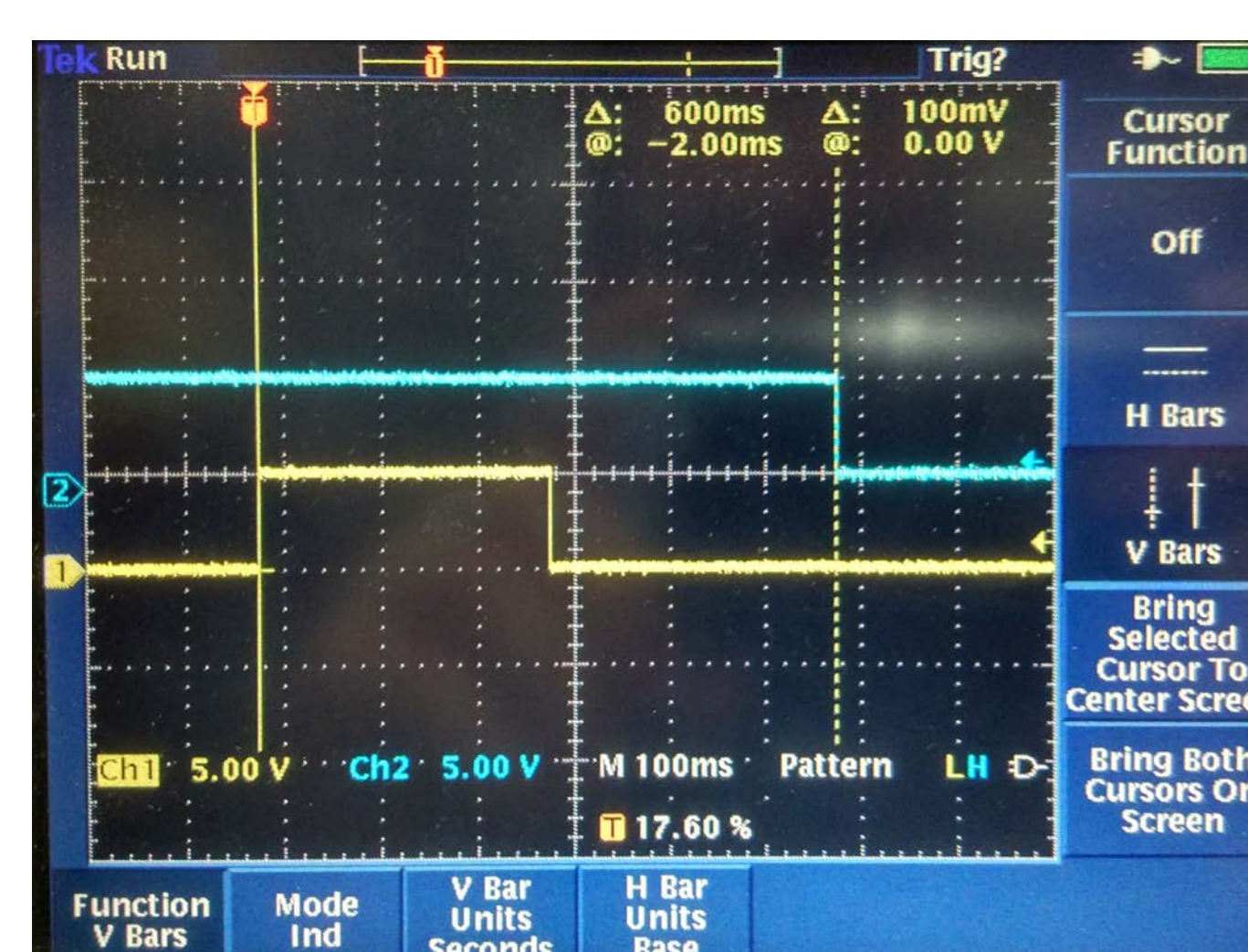


Fig 6: SH and Protection pulses

After Cold Powering tests of Feather M0.4 (beginning October 2016) I started developing the new software version for the cRIO DAQ system. By the moment I've made the next improvements in the software: Simplification and reorganisation of the software structure; Solved the oversizing sample rate problem, found during Run 3, having now a single one acquisition sample rate at 1kHz; Solved the SH pulses width problem (Fig. 6), found during Run 4.

At the present, are operative in the software the features of data acquisition, data storage in the HD and the SH Firing System. On the other hand I'm currently working in the Quench Detection system implementation.

Acronyms & Definitions

1. HTS = High Temperature Superconductor
2. DAQ system = Data Acquisition system
3. Quench = Transition from superconducting to resistive state that takes place in supeconducting magnets.
4. SH = Spot Heater
5. NI = National Instruments
6. FPGA = Field Programmable Gate Array
7. RT = Real Time

References

- [1] K. A. Broekens, "Quench Detection Study for a HTS Model Dipole Magnet", October 2016
- [2] G. Kirby; J. van Nugteren; H. Bajas; V. Benda; A. Ballarino; M. Bajko; L. Bottura; K. Broekens; M. Canale; A. Chiuchiolo; L. Gentini; N. Peray; J. Perez; G. de Rijk; A. Rijllart; L. Rossi; J. Murtomaeki; J. Mazet; F. Pincot; G. Volpini; M. Durante; P. Fazilleau; C. Lorin; A. Stenvall; W. Goldacker; A. Kario; A. Usoskin, "First Cold Powering Test of REBCO Roebel Wound Coil for the EuCARD2 Future Magnet Development Project," in IEEE Transactions on Applied Superconductivity, vol. PP, no.99, pp.1-1
- [3] G. Kirby; H. Bajas; A. Chiuchiolo; F. Pincot; A. Diaz, "Feather_0.4 4-run test report", 5th October 2016.
- [4] National Instruments, "Compactrio developers guide", <https://www.ni.com/compactriodevguidel>.